

Bridge Management System

A unified digital platform for managing Australia's bridge assets — from inspection to NHVR compliance.

5,000+ Bridges

8 States

10 Apps



Built on SAP BTP

Cloud Foundry Runtime
SAP HANA Cloud
Fiori / UI5 Frontend



Data Fragmented Across Systems

Bridge records, inspection reports, load ratings, and NHVR permits stored in separate spreadsheets, emails, and legacy systems with no single source of truth.



No Real-Time Visibility

State transport agencies and NHVR officers cannot see current bridge condition, active restrictions, or permit status without manual requests and phone calls.



Compliance Risk

Overloaded heavy vehicles crossing undercapacity bridges risk structural damage and public safety. Manual gazette processes delay approval and create legal exposure.



BMS solves all three — one platform, real-time data, role-based access, built on enterprise-grade SAP BTP infrastructure.

What BMS Does — Capability Map

Transport for NSW



Asset Registry



Engineers / Managers

Centralised record of every bridge: location, structural type, materials, load ratings, span details, condition history and photos.



GIS Map & Spatial View



All staff / field teams

Interactive map showing condition heat-maps, active weight restrictions, NHVR routes, and real-time viewport data for any zoom level.



Inspection & Condition



Field Inspectors

Capture structured condition ratings in the field. Attach photos, log defects, track structural elements. Trend charts show deterioration over time.



NHVR Compliance



NHVR Officers / Managers

Heavy-vehicle route assessment against bridge capacity. Gazette expiry tracking. Class 1–3 permit approvals with full audit trail.



Bulk Data Management



Administrators

Excel/CSV mass upload and edit for thousands of bridges. Validation preview before commit. Supports all dataset types.



Analytics & Reporting



Executives / Planners

Network KPIs, condition distribution charts, restriction trends, NHVR gazette expiry timeline. Exportable bridge cards and reports.

Who Uses BMS — Roles & Business Value

Transport for NSW



BMS Administrator

IT / Asset Management

System configuration, audit log review, bulk data upload, user access management.



Bridge Manager

State Transport Agency

Create and maintain bridge records, manage weight restrictions, oversee condition capture programme.



BMS Operator

TfNSW / State Roads

NHVR compliance checks, mass data edits, map-based spatial queries, gazette processing.



Inspector

Field Engineering Teams

Record condition ratings on-site, log defects, capture photos, track structural element status.



Viewer / Planner

Policy / Executive teams

Read-only access to maps, reports, dashboards, and KPI snapshots for planning and strategy.

Standard	Domain	How BMS Meets It
ISO 55000/55001	Asset Management	Lifecycle records, condition trends, criticality scoring, and maintenance planning data all captured in BMS.
AS 5100 (Bridge Design)	Structural Adequacy	Load rating fields, structural adequacy rating, and span data align with AS 5100 bridge assessment methodology.
NHVR Act 2012	Heavy Vehicle Compliance	BMS stores gazette data, route assessments, and permit approvals meeting NHVR regulatory requirements.
ISO/IEC 27001	Information Security	XSUAA role-based access, row-level state isolation, CSRF protection, and audit logging on all mutations.
SOC 2 (SAP BTP)	Cloud Security	SAP BTP Cloud Foundry and HANA Cloud hold SOC 2 Type II certification — inherited by BMS workloads.
WCAG 2.1 AA	Accessibility	SAP Fiori / UI5 framework provides accessibility compliance for all BMS screen-based apps.

✅ BMS is built on SAP BTP — a globally certified enterprise cloud platform. All data remains within Australian data centres.

Single

Source of Truth

No more conflicting spreadsheets

Real-time

Condition Data

Field → system in minutes

Automated

NHVR Compliance

Gazette expiry alerts

Audit-ready

Change Logs

Every mutation recorded



One platform for all bridge data



Role-based access — right people, right data



Bulk upload replaces manual data entry



Multi-state with row-level state isolation



GIS-first: every record on the map



Trend analysis catches deterioration early



Automated alerts for expiry and condition drops



Compliance-ready audit trail for ISO 55001

Next Steps

01

UAT Sign-off

Complete user acceptance testing across all 5 role profiles. Target: all critical paths green.

02

Data Migration

Finalise bulk upload of existing bridge records from legacy systems using Excel import.

03

BTP Production Deploy

Run `npm run validate-build` → cf deploy to production CF space. Smoke test /health endpoint.

04

Training & Onboarding

Role-based training sessions for Inspectors, Operators, and NHVR Officers.

05

Hypercare Period

4-week hypercare with monitoring via Application Logging and weekly KPI reviews.



Contact

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[SAP BTP Documentation](#)

help.sap.com/btp